

Math 428, Spring 2024, Homework 5 Solutions

Section 5.1: 6

Solution.

$$\cot^2(z) = \frac{\cos(z)^2}{\sin(z)^2} = \frac{1}{z^2} \frac{\left(1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \dots\right)}{\left(1 - \frac{1}{3!}z^2 + \frac{1}{5!}z^4 - \dots\right)}$$

Let $g(z) = \frac{\left(1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \dots\right)}{\left(1 - \frac{1}{3!}z^2 + \frac{1}{5!}z^4 - \dots\right)}$. Then

$$\text{Res}(\cot^2(z), 0) = g'(0) = 0,$$

where the last can be seen by using the quotient rule, or from the fact that $g(z)$ is even in z .

Section 5.1: 8

Solution.

$$\frac{1}{z^2 \sin z} = \frac{1}{z^3(1 - z^2/3! + z^4/5! - \dots)}$$

Pole order 3, so do long division on $1/(1 - z^2/3! + z^4/5! - \dots)$, keeping only terms up to order z^2 .

$$\left(1 - \frac{1}{6}z^2\right) \overline{1} = 1 + \frac{1}{6}z^2 + \dots$$

Laurent expansion is

$$\frac{1}{z^3}(1 + \frac{1}{6}z^2 + \dots) = \frac{1}{z^3} + \frac{1}{6} \frac{1}{z} + \dots$$

$$\text{So: } \text{Res}\left(\frac{1}{z^2 \sin z}, 0\right) = \frac{1}{6}$$

Section 5.1: 10

Solution. $\sinh z = z + \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \dots$, so

$$\frac{\sinh z}{z^4} = \frac{1}{z^3} + \frac{1}{3!} \frac{1}{z} + \frac{1}{5!}z - \dots$$

So: $\operatorname{Res}\left(\frac{\sinh z}{z^4}, 0\right) = \frac{1}{3!} = \frac{1}{6}$

Section 5.2: 1

Solution.

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{5 - 4\cos\theta} &= -i \int_{|z|=1} \frac{1}{5 - 2(z + z^{-1})} \frac{dz}{z} = \frac{i}{2} \int_{|z|=1} \frac{dz}{z^2 - \frac{5}{2}z + 1} \\ &= \frac{i}{2} \int_{|z|=1} \frac{dz}{(z-2)(z-\frac{1}{2})} = (2\pi i) \frac{i}{2} \operatorname{Res}\left(\frac{1}{(z-2)(z-\frac{1}{2})}, \frac{1}{2}\right) = \frac{2\pi}{3} \end{aligned}$$

Section 5.2: 2

Solution.

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{10 + 6\sin\theta} &= -i \int_{|z|=1} \frac{1}{10 + 3(z - z^{-1})/i} \frac{dz}{z} = \frac{1}{3} \int_{|z|=1} \frac{1}{z^2 + \frac{10}{3}iz - 1} \\ &= \frac{1}{3} \int_{|z|=1} \frac{dz}{(z+3i)(z+\frac{1}{3}i)} = (2\pi i) \frac{1}{3} \operatorname{Res}\left(\frac{1}{(z+3i)(z+\frac{1}{3}i)}, -\frac{1}{3}i\right) = \frac{\pi}{4} \end{aligned}$$

Additional problem 1.

Solution. To evaluate the limit as $y \rightarrow +\infty$, write

$$\tan z = \frac{1}{i} \frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}} = -i \frac{e^{2iz} - 1}{e^{2iz} + 1} = i \frac{1 - e^{2iz}}{1 + e^{2iz}}$$

If $z = x + iy$, then $|e^{2iz}| = |e^{2ix}e^{-2y}| = e^{-2y}$, so $\lim_{y \rightarrow +\infty} e^{2iz} = 0$. Thus,

$$\lim_{y \rightarrow +\infty} i \frac{1 - e^{2iz}}{1 + e^{2iz}} = i \frac{1 - 0}{1 + 0} = i.$$

To evaluate the limit as $y \rightarrow -\infty$, write

$$\tan z = \frac{1}{i} \frac{e^{iz} - e^{-iz}}{e^{iz} + e^{-iz}} = -i \frac{1 - e^{-2iz}}{1 + e^{-2iz}}$$

If $z = x + iy$, then $|e^{-2iz}| = |e^{-2ix}e^{2y}| = e^{2y}$, so $\lim_{y \rightarrow -\infty} e^{-2iz} = 0$. Thus,

$$\lim_{y \rightarrow -\infty} -i \frac{1 - e^{-2iz}}{1 + e^{-2iz}} = -i \frac{1 - 0}{1 + 0} = -i.$$

Additional problem 2.

Solution. $f(z)$ is analytic in $z : (z-1)/(z+1) \in \mathbb{C} \setminus (-\infty, 0]$. In order to be defined we must have $z \neq -1$.

If we let $w = (z-1)/(z+1)$, then $z = (1+w)/(1-w)$. We need to exclude the set of z for which $w \in (-\infty, 0]$. Since this means w is real we see that the excluded set of z is real. An inspection of the graph shows that if $w \in (-\infty, 0]$ then we have $z \in (-1, 1]$. Thus, if $z \in \mathbb{C} \setminus [-1, 1]$ then $w \in \mathbb{C} \setminus (-\infty, 0]$, and the function is analytic there.

We can calculate the derivative for $z \in \mathbb{C} \setminus [-1, 1]$ using the chain rule and fraction rule:

$$\begin{aligned} \frac{1}{2} \log\left(\frac{z-1}{z+1}\right) &= \frac{1}{2} \frac{1}{\frac{z-1}{z+1}} \left(\frac{z-1}{z+1}\right)' \\ &= \frac{1}{2} \frac{z+1}{z-1} \left(\frac{1}{z+1} - \frac{z-1}{(z+1)^2}\right) \\ &= \frac{1}{2} \left(\frac{1}{z-1} - \frac{1}{z+1}\right) \\ &= \frac{1}{z^2-1} \end{aligned}$$